

## a Case 127: three different mappings, the same minimum event count

140-reaction coordinate set · GRAFT: 3 classes · SLAP sweep: 3 · Native SLAP: 1

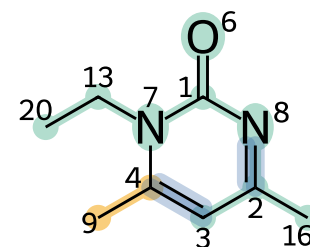
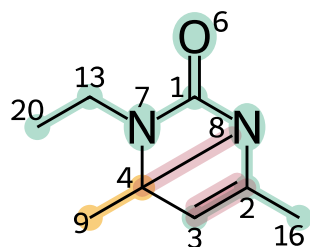
● ● ● Matched fragments keep their colors

— Break / weaken

— Form / strengthen

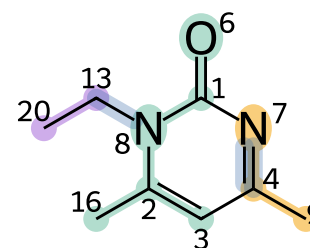
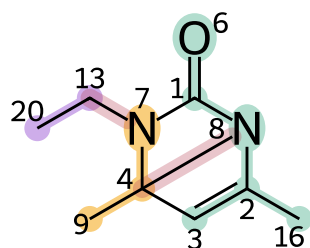
### A 4 events

2 negative  
2 positive



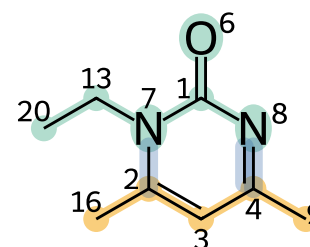
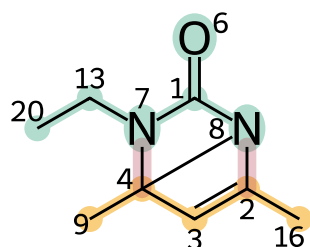
### B 4 events

2 negative  
2 positive



### C 4 events

2 negative  
2 positive



Atom numbers track reactant identities. Colors are assigned within each row; spectator H atoms are implicit.

## b Allowed symmetry remains inside a candidate

### Candidate A

Methyl C<sub>9</sub>:  
exchange H<sub>10</sub> and H<sub>11</sub>



**Certified allowed**

Same event class

All other assignments fixed

R = the rest of the molecule. This spectator-H example illustrates the conditional symmetry query.